An explicit constant exceeding $(3-\sqrt{5})/2$ for union-closed families: a machine-checked certificate with independent replay

The union-closed certification project*

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Abstract

Let

$$\psi = \frac{3-\sqrt{5}}{2} \quad \text{and} \quad t_{\mathrm{cert}} = \frac{955165028125263}{2500000000000000} = 0.3820660112501052.$$

We establish the first explicit union-closed frequency constant strictly above ψ with a complete proof chain: the exact rational comparison $t_{\mathrm{cert}} \geq \psi + 10^{-4}$ and the required five-parameter functional inequality are PROVED by machine-checked artifacts, while the single entropy-to-union-closed implication is CITED from Cambie [5, Question 2 and §4]. Thus every finite union-closed family $\mathcal{F} \neq \{\emptyset\}$ has an element in at least $t_{\mathrm{cert}}|\mathcal{F}|$ member sets, subject precisely to that cited implication. The new proof decomposes the OR-entropy kernel into one positive square minus an explicit positive-semidefinite defect, reduces the global minimum to two pair-orbits (five parameters), and certifies nonnegativity by an Arb interval branch-and-bound. A frozen collector independently replayed all eight one-byte-per-node proof traces and printed `COMPOSITE CERTIFICATE`. As a second, independent contribution, we prove Liu's previously only numerically verified positive-semidefiniteness hypothesis behind the best published constant, 0.382709087918741; his separate nine-parameter global-optimization hypothesis remains open, so that constant is still conditional. We do not claim Frankl's constant $1/2$, optimality of t_{cert}, certification of the reported $0.3823455\ldots$ or $0.3827090\ldots$ constants, or a reproof of the one cited implication.

*Repository artifact: `math/uc/` of the `jinleic-workspace` tree; every claim carries a machine-checked artifact named in the text.

1 Introduction

1.1 Frankl's conjecture and the pre-constant era

A finite family of sets is *union-closed* if $A \cup B$ belongs to the family whenever A and B do. The union-closed sets conjecture says that every finite union-closed family $\mathcal{F} \neq \{\emptyset\}$ contains an element in at least $|\mathcal{F}|/2$ member sets. It is commonly attributed to Péter Frankl in 1979, although the historical record describes it as a folklore conjecture already known in the 1970s [4, §1]. We therefore use "Frankl's conjecture" as the conventional name, not as a claim of sole priority.

Before the 2022 entropy breakthrough, the general quantitative bounds decayed with the size of the family. Knill [9] and Wójcik [18] obtained frequency fractions of order $\Omega(1/\log|\mathcal{F}|)$; in particular these were not positive constants independent of $|\mathcal{F}|$. The relevant early author is Emanuel *Knill*.

1.2 The entropy ladder: 1/100 and the golden-ratio barrier

Gilmer [7] introduced the entropy method that produced the first dimension-free constant, proving $1/100$. For a regular Borel probability law μ on $[0, 1]$, its iid OR-entropy and one-coordinate entropy are

$$\mathrm{Q}(\mu) = \iint h(p + q - pq)\, d\mu(p)\, d\mu(q), \qquad \mathrm{L}(\mu) = \int h(p)\, d\mu(p),$$

where $h(u) = -u \log_2 u - (1-u) \log_2(1-u)$, continuously extended by $h(0) = h(1) = 0$. The sharp iid inequality was then established independently in near-simultaneous work of Alweiss–Huang–Sellke [1], Chase–Lovett [6], Sawin [14], and Pebody [12]. It yields

$$\psi = \frac{3 - \sqrt{5}}{2} \approx 0.3819660112501051,$$

the largest explicit constant above which no complete proof had previously been available. Chase–Lovett also showed that ψ is sharp for their approximate-union-closed formulation, and it is the sharp barrier for the iid entropy strategy.

Sawin's revised paper [14, §2] does more than prove the ψ bound: it sketches how a convex combination of the iid coupling and a dependent max-entropy coupling gives some unspecified constant strictly larger than ψ. The sketch deliberately does not calculate an explicit improvement. This is the conceptual second strategy used by Yu, Cambie, and the present work.

1.3 Claims beyond the golden-ratio barrier and their proof status

The distinction among PROVED, CITED, and NUMERICAL statements is load-bearing here.

Yu. Yu [19, §4] claims the explicit value $0.38234\ldots$ (arXiv:2212.00658v2). The argument invokes Krein–Milman/Carathéodory and then a concavity inequality to pass from extreme points to a finite-support optimization. That inference is false on Yu's own extreme-point class. The repository artifact `uc/yu_gap.py` gives a particularly simple structured counterexample: with $Q_1 = \delta_{(0.15,0.15)}$, $Q_2 = (1-\beta)\delta_{(0.10,0.10)} + \beta Q_{1,1}$ at mean 0.382345533366702721, and mixing weight 1/2, it reports the claimed concavity defect as -2.476271×10^{-2}. This displayed sign is a NUMERICAL diagnostic, not a proof step in our theorem; it refutes the specific reduction with a margin far from roundoff. Theorem 3.2 below supplies a different support reduction with all hypotheses retained.

Cambie. Cambie [5] formulates the two-strategy functional inequality as Question 2 and reports

$$c^\star \approx 0.3823455333667027.$$

Version 2 explicitly describes its finite-dimensional verification as follows: *"In contrast to the detailed work in e.g. [1] for the constant* $(3-\sqrt{5})/2$*, this is done slightly less rigorous."* In other words, that optimization was carried out slightly less rigorously: the paper itself notes local-minimum and finite-precision issues. We do not use that numerical verification. We use only the separate implication in Cambie's §4 that converts a valid answer to Question 2 into the union-closed frequency bound; that one imported step is always labeled CITED below.

Liu. Liu [10] develops conditionally iid couplings. The explicit constant in Theorem 13,

$$0.382709087918741,$$

is conditional on two hypotheses stated there: the positive-semidefiniteness hypothesis of §V-A and the global-minimizer-structure hypothesis of §V-B. Both were reported as numerically verified rather than proved. As a second, independent contribution, Theorem 6.1 proves the §V-A hypothesis. The nine-parameter §V-B hypothesis remains open, so Liu's decimal is still conditional and is not a theorem claimed here. Related entropy developments include Wakhare [16] and Lu–Raz [11].

Quantitative ladder.

- PROVED (size-dependent): $\Omega(1/\log|\mathcal{F}|)$, by Knill [9] and Wójcik [18].
- PROVED: $1/100$, by Gilmer [7].
- PROVED: $\psi = (3-\sqrt{5})/2$, by Alweiss–Huang–Sellke [1], Chase–Lovett [6], Sawin [14], and Pebody [12].
- Claimed with a finite-dimensional proof gap: $c^\star \approx 0.3823455333667027$, by Yu [19] and Cambie [5].
- Conditional because the nine-parameter Hypothesis 2 remains open (Hypothesis 1 is proved in this work): 0.382709087918741, in Liu's Theorem 13 [10].
- PROVED functional inequality plus one CITED implication: $t_{\mathrm{cert}} = 0.3820660112501052$, in this work and Cambie's §4 [5].

The last item is the exact rational claim $t_{\mathrm{cert}} \geq \psi + 10^{-4}$, never an equality between the rational target and the irrational number $\psi + 10^{-4}$.

1.4 The result proved here

For a compact metric space X, let $\mathcal{M}(X)$ be the real vector space of finite signed regular Borel measures, $\mathcal{M}_+(X)$ its positive cone, and

$$\mathcal{P}(X) = \{\mu \in \mathcal{M}_+(X) : \mu(X) = 1\}.$$

All measures and couplings in the theorem statements are regular Borel measures. For $\mu \in \mathcal{P}([0,1])$ put $\mathrm{mean}(\mu) = \int_0^1 p\, d\mu(p)$ and $\mathrm{B}(\mu) = 1 - \mathrm{mean}(\mu)$. A self-coupling of μ means an element of $\mathcal{P}([0,1]^2)$ whose two marginals both equal μ; the set of all such couplings is denoted $\Pi(\mu,\mu)$. Weak-$*$ topology is used only where it is named explicitly.

Set

$$\begin{aligned}\alpha_0 &= \frac{356069}{10000000} = 0.0356069,\\ t_{\mathrm{cert}} &= \frac{955165028125263}{2500000000000000}\\ &= 0.3820660112501052.\end{aligned}$$

The rational inequality $t_{\mathrm{cert}} \geq \psi + 10^{-4}$ is checked exactly by the campaign preflight.

Theorem 1.1 (Machine-checked functional inequality)**.** *Let* $\alpha_0 = 0.0356069$ *and let* $t_{\mathrm{cert}} = 0.3820660112501052$ *denote the exact rationals displayed above. For every* $\mu \in \mathcal{P}([0,1])$ *satisfying* $\mathrm{mean}(\mu) \leq t_{\mathrm{cert}}$*, define*

$$\begin{aligned} \mathrm{L}(\mu) &= \int h(p)\, d\mu(p), \\ \mathrm{Q}(\mu) &= \iint h(p+q-pq)\, d\mu(p)\, d\mu(q), \\ \mathrm{C}(\mu) &= \min_{\pi \in \Pi(\mu,\mu)} \iint h\big(s^\star(p,r)\big)\, d\pi(p,r), \\ s^\star(p,r) &= \max\big(p, r, \min(p+r, \tfrac{1}{2})\big). \end{aligned}$$

Then

$$\mathrm{F}(\mu) := (1-\alpha_0)\mathrm{Q}(\mu) + \alpha_0 \mathrm{C}(\mu) - \mathrm{L}(\mu) \geq 0. \tag{1}$$

Corollary 1.2 (Union-closed consequence; one cited implication)**.** *Assume the entropy-to-union-closed implication in Cambie's Question 2 and §4 [5]: for* $\alpha \in [0,1]$ *and* $t \in [0,1]$*, validity of the corresponding functional inequality for every* $\mu \in \mathcal{P}([0,1])$ *with* $\mathrm{mean}(\mu) \leq t$ *implies the frequency bound* t *for every finite union-closed family other than* $\{\emptyset\}$*. Then every finite union-closed family* $\mathcal{F} \neq \{\emptyset\}$ *has an element belonging to at least*

$$t_{\mathrm{cert}}|\mathcal{F}| = 0.3820660112501052\,|\mathcal{F}|$$

member sets.

Theorem 1.1 is PROVED here. The implication explicitly assumed in Corollary 1.2 is CITED and is not re-proved. Together they give the first explicit constant strictly above ψ with a complete proof chain. The scope is exact: we do not prove Frankl's 1/2, certify $c^\star$, certify Liu's conditional decimal, prove optimality, or claim a result at any larger target.

The proof has four structural ingredients:

- (A) an exact OR-entropy-kernel decomposition; see §2 and `decomposition.py`;
- (A′) the identity $\mathrm{Q} = 2\mathrm{BL} - \mathrm{E}$ with an explicit positive-semidefinite defect; see §2 and `reduction.py`;

(B‴) a support reduction to at most two pair-orbits, hence five parameters; see §3 and `thmB3_proof.py`;

1. a scale-free margin lemma and the rh bound controlling the entropy sinks; see §4, `margin_lemma.py`, and `lemma_rh_proof.py`.

An Arb interval branch-and-bound then certifies the reduced functional (§5). Campaign I committed all eight slices, totaling 488,465,854 boxes with zero residual and empty final stacks. Its frozen collector replayed every trace and printed `COMPOSITE CERTIFICATE` on 2026-08-21. Appendix A gives the exact replay command and trust base. Every search heuristic or sampled diagnostic is labeled NUMERICAL and is not a proof step.

2 The exact scaffold: Theorems A and A-prime

Throughout, logarithms are natural unless the subscript 2 shows an explicit base. We use the natural-log kernels

$$H(u) = -u \ln u - (1-u)\ln(1-u), \qquad T(u) = u + (1-u)\ln(1-u), \tag{2}$$

extended continuously by $H(0) = 0$, $T(0) = 0$, $T(1) = 1$; recall that $h(u) = H(u)/\ln 2$.

Lemma 2.1 (T-series)**.** *For $0 \le u \le 1$,*

$$T(u) = \sum_{n=2}^{\infty} \frac{u^n}{n(n-1)}, \qquad T(1) = \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = 1. \tag{3}$$

Proof. $T'(u) = -\ln(1-u) = \sum_{m\ge1} u^m/m$ on $[0,1)$, so integrating from 0 gives $T(u) = \sum_{m\ge1} u^{m+1}/(m(m+1)) = \sum_{n\ge2} u^n/(n(n-1))$; the endpoint follows by Abel's theorem or by the telescoping $1/(n(n-1)) = 1/(n-1) - 1/n$. The machine artifact `decomposition.py` checks the closed form against the series at five points to 10^{-30} and the telescoping sum exactly. □

Theorem 2.2 (Theorem A; `decomposition.py`)**.** *Let H, T be the continuous kernels in (2), and set $g(x) = x(1 - \ln x)$ and $a(x) = -x \ln x$, with $g(0) = a(0) = 0$. For every $x, y \in [0,1]$,*

$$H(xy) = g(x)\,g(y) - a(x)\,a(y) - T(xy). \tag{4}$$

Moreover $T(xy) = \sum_{n\ge2}(xy)^n/(n(n-1))$ is positive semidefinite in the following explicit measure-cone sense:

$$\iint T(xy)\,d\eta(x)\,d\eta(y) \ge 0 \qquad \textit{for every } \eta \in \mathcal{M}([0,1]).$$

Thus (4) *writes the OR-entropy kernel as the single positive square* $g \otimes g$ *minus the positive-semidefinite remainder* $a \otimes a + T$*.*

Proof. Since $g = a + \mathrm{id}$,

$$g(x)g(y) - a(x)a(y) = x\,a(y) + y\,a(x) + xy = -xy\ln(xy) + xy = H(xy) + T(xy),$$

the last step by $H(u) + T(u) = -u\ln u + u$ from (2). Rearranging gives (4). The PROVED gate in `decomposition.py` asserts the identity as an exact SymPy tautology on three symbolic atoms. Its checks on 200 random signed measures and on a 200×200 grid are additional NUMERICAL controls, not proof steps. $\square$

Remark 2.3. The "at most one positive square" statement is the algebraic heart of the method: it says the two-strategy entropy Q differs from a square of a linear functional by a positive-semidefinite remainder, so that a variance ("defect") term appears with a sign we can control. This is the property that makes the later support reduction possible; see §3.

We now pass to the probability-theoretic formulation. For $\mu \in \mathcal{P}([0,1])$ write, with the substitution $x = 1 - p$ in the W-kernel,

$$\begin{aligned} \mathrm{E}(\mu) &= \frac{1}{\ln 2} \iint W(1-p,\, 1-q)\, d\mu(p)\, d\mu(q), \\ W(x,y) &= xy - x\,T(y) - y\,T(x) + T(xy). \end{aligned} \tag{5}$$

Theorem 2.4 (Theorem A′; `reduction.py`)**.** *Let* W *and* E *be as in* (5). *For every* $x, y \in [0,1]$,

$$W(x,y) = \sum_{n=2}^{\infty} \frac{(x - x^n)(y - y^n)}{n(n-1)}. \tag{6}$$

Consequently W *is positive semidefinite on the full signed-measure space:*

$$\iint W(x,y)\, d\eta(x)\, d\eta(y) \geq 0 \qquad \textit{for every } \eta \in \mathcal{M}([0,1]).$$

For every $\mu \in \mathcal{P}([0,1])$*, with* $\widetilde{\mu} = (p \mapsto 1-p)_{\#}\mu$ *and* $M_n = \int x^n\, d\widetilde{\mu}(x)$*, one has*

$$\mathrm{Q}(\mu) = 2\,\mathrm{B}(\mu)\,\mathrm{L}(\mu) - \mathrm{E}(\mu), \tag{7}$$

$$\ln 2 \cdot \mathrm{E}(\mu) = \sum_{n=2}^{\infty} \frac{(M_1 - M_n)^2}{n(n-1)}, \tag{8}$$

$$\frac{W(x,x)}{\ln 2} = 2x\,h(x) - h(x^2) \qquad (0 \leq x \leq 1). \tag{9}$$

In particular $\mathrm{E}(\mu) \geq 0$ *for every* $\mu \in \mathcal{P}([0,1])$.

Proof. For (6), expand termwise over the four convergent series of (5) using (3): the telescoping identity $\sum_{n\geq 2} 1/(n(n-1)) = 1$ and the two evaluations $\sum_{n\geq 2} xy^n/(n(n-1)) = x\,T(y)$ and its symmetric partner reproduce W. Each summand $(x - x^n)(y - y^n)$ is a rank-one kernel, so W is a sum of nonnegative squares; its quadratic form against a finite signed measure is therefore a sum of squares, giving (8) with $\int (x - x^n)\, d\widetilde{\mu} = M_1 - M_n$.

For (7), note that $p + q - pq = 1 - xy$ with $x = 1 - p$, $y = 1 - q$, so $h(p + q - pq) = H(xy)/\ln 2$, and by Theorem 2.2

$$\ln 2 \cdot \mathrm{Q} = \Big(\int g\, d\widetilde{\mu}\Big)^2 - \Big(\int a\, d\widetilde{\mu}\Big)^2 - \iint T(xy)\, d\widetilde{\mu}\, d\widetilde{\mu}.$$

Let $\overline{T} = \int T\, d\widetilde{\mu}$ and $B = \int x\, d\widetilde{\mu}$; then $\int g\, d\widetilde{\mu} = \int a\, d\widetilde{\mu} + B$, and from $H(x) = a(x) + x - T(x)$,

$$\ln 2 \cdot \mathrm{L} = \int a\, d\widetilde{\mu} + B - \overline{T}.$$

A direct expansion gives

$$\ln 2\,(2B\,\mathrm{L} - \mathrm{Q}) = B^2 - 2B\,\overline{T} + \iint T(xy)\, d\widetilde{\mu}\, d\widetilde{\mu} = \iint W\, d\widetilde{\mu}\, d\widetilde{\mu} = \ln 2 \cdot \mathrm{E},$$

which is (7). Finally (9) follows from the identity $T(u) = -u \ln u + u - H(u)$: $W(x, x) = x^2 - 2x\,T(x) + T(x^2) = 2x\,H(x) - H(x^2)$. The PROVED gate in `reduction.py` verifies (6) against the closed form within an analytic tail bracket (the remainder is at most $1/N$ after N terms) and asserts the symbolic identity chain. Its random-law evaluations and four-point evaluations, including the golden point, are NUMERICAL controls only. ☐

Substituting (7) into (1) yields the form used throughout:

Corollary 2.5 (Defect form of the functional)**.** *For every $\alpha \in [0, 1]$ and every $\mu \in \mathcal{P}([0, 1])$, with $\mathrm{F}(\mu) = (1 - \alpha)\mathrm{Q}(\mu) + \alpha \mathrm{C}(\mu) - \mathrm{L}(\mu)$,*

$$\mathrm{F}(\mu) = \big[2(1 - \alpha)\,\mathrm{B}(\mu) - 1\big]\,\mathrm{L}(\mu) + \alpha\,\mathrm{C}(\mu) - (1 - \alpha)\,\mathrm{E}(\mu). \tag{10}$$

2.1 The iid golden point and the golden-ratio barrier

For $\alpha \in [0, 1]$ and a point mass $\mu = \delta_p$ with $p \in [0, 1]$, the only self-coupling is $\delta_{(p,p)}$, and

$$\mathrm{F}(\delta_p) = \begin{cases} h\big((1 - p)^2\big) - h(p), & \alpha = 0, \\ (1 - \alpha)\, h\big((1 - p)^2\big) + \alpha\, h\big(s^\star(p, p)\big) - h(p), & \alpha \geq 0, \end{cases} \tag{11}$$

since $\mathrm{Q}(\delta_p) = h(2p - p^2) = h(1 - (1 - p)^2)$ and $\mathrm{C}(\delta_p) = h(s^\star(p, p))$.

Corollary 2.6 (The iid point-mass barrier; `reduction.py`). *For every $p \in [0,1]$, at $\alpha = 0$,*

$$\mathrm{F}(\delta_p) \geq 0 \iff (1-p)^2 \geq p \iff p \leq \psi = \frac{3-\sqrt{5}}{2}.$$

For $p \in [0, \frac{1}{2}]$, both $(1-p)^2$ and p lie in $[0, \frac{1}{2}]$ whenever the inequality can change sign, and h is increasing there; for $p > \frac{1}{2}$, one has $h((1-p)^2) < h(1-p) = h(p)$. The quadratic inequality has smaller root ψ. At the golden point $x = 1/\varphi$, $x^2 = 1 - x$ exactly, so F vanishes: ψ is precisely where the PSD defect E first overtakes $(2\mathrm{B}-1)\mathrm{L}$.

For later comparisons, define the two-orbit obstruction constants without a decimal approximation. Let $b^\star$ be the larger root in $(0, \frac{1}{2})$ of

$$h(b)\,[2 - h(b)] = h\big((1-b)^2\big),$$

put $a^\star = [1 - h(b^\star)]/[2 - h(b^\star)]$, and set $c^\star = a^\star + (1 - a^\star)b^\star$.

The endpoint value $\mathrm{F}(\delta_\psi) = \alpha[1 - h(\psi)] \approx 0.0405813\,\alpha > 0$ for $\alpha > 0$, so point masses do not cap the constant once the second strategy is active. The crossover value is

$$\alpha_{\mathrm{crit}} = \frac{h(c^\star) - h\big((1-c^\star)^2\big)}{1 - h\big((1-c^\star)^2\big)} \approx 0.0144054585140081. \tag{12}$$

Lemma 2.7 (Point-mass threshold; `reduction.py` §6). *Let $\alpha \in [0,1)$, let $\mathrm{F}(\delta_p) = (1-\alpha)h(2p-p^2) + \alpha h(s^\star(p,p)) - h(p)$ for $p \in [\frac{1}{4}, \frac{1}{2}]$, and let $c^\star, \alpha_{\mathrm{crit}}$ be defined above. The equation $\mathrm{F}(\delta_p) = 0$ has a unique root in $(\frac{1}{4}, \frac{1}{2})$, and*

$$\textit{that root exceeds } c^\star \iff \alpha > \alpha_{\mathrm{crit}}.$$

For $\alpha = \alpha_0 = 0.0356069 > \alpha_{\mathrm{crit}}$, point masses are therefore not binding.

Proof sketch; complete in `reduction.py`. On $(1/4, 1/2)$, $s^\star(p,p) = 1/2$ is *constant* (since $p \leq 1/2 \leq p + p$), so the α-term drops out of $\partial_p \mathrm{F}(\delta_p)$, giving

$$\frac{d}{dp}\,\mathrm{F}(\delta_p) = (1-\alpha)(2-2p)\,h'\big(2p - p^2\big) - h'(p).$$

On $[p_0, \frac{1}{2})$ with $p_0 = 1 - 1/\sqrt{2}$ where $2p - p^2 \geq \frac{1}{2}$, each factor has a proved sign: $h'(2p - p^2) \leq 0$, $2 - 2p > 0$, $h'(p) > 0$, so the derivative is strictly negative. On $[0, p_0]$, both $h(2p - p^2) \geq h(p)$ and $h(s^\star(p,p)) \geq h(p)$ hold uniformly, so $\mathrm{F} \geq 0$ for *every* $\alpha \in [0,1]$; and $\mathrm{F}(\delta_{1/2}) < 0$ for $\alpha < 1$. Hence there is a unique root. Its position relative to $c^\star$ is decided by the sign at $c^\star$, which is affine in α and vanishes exactly at (12). The artifact asserts this identity chain. Its bisection over a grid of 401 points per α is an additional NUMERICAL control, not the assertion. □

2.2 Why the variance term is load-bearing

Dropping the variance in the crudest way — replacing E by a pointwise upper bound derived from (9) — gives a bound whose ratio form is governed by $\kappa^\star = \max_{x\in(0,1)}[2x - h(x^2)/h(x)]$, i.e. an explicit constant $(1-\kappa^\star)/2 \approx 0.34753 < \psi$ (asserted in `reduction.py` §8). The maximizer sits at $p^\star \approx 0.117$, far below the mean bound, so the pointwise route *wastes* the constraint $\mathbb{E}p \le t$: the variance term is load-bearing, and one must either retain it (as we do, via the support reduction and the certificate) or control it sharply enough to win it back.

3 Support reduction: Theorem B-triple-prime

The defect term E is convex in μ (a positive semidefinite quadratic form), so the certificate cannot simply minimize a concave relaxation; instead we prove that the global minimum of the exact functional F is attained on a tiny parameter family. The functional must first be recast on *pair-orbit measures.*

Definition 3.1 (Pair-orbit measures)**.** Let

$$\Delta = \{(p,q) \in [0,1]^2 : p \le q\},$$

and let $\mathcal{P}(\Delta)$ denote the regular Borel probability measures on this compact triangle. A point $(p,q) \in \Delta$ is a *pair orbit.* Every $\nu \in \mathcal{P}(\Delta)$ induces the marginal law

$$\mu_\nu = M(\nu) = \frac{(\pi_1)_\#\nu + (\pi_2)_\#\nu}{2} \in \mathcal{P}([0,1]),$$
$$\int f\,d\mu_\nu = \int_\Delta \frac{f(p)+f(q)}{2}\,d\nu(p,q) \qquad (f \in C[0,1]).$$

A *two-orbit law* is

$$\nu = w\delta_{(p_1,q_1)} + (1-w)\delta_{(p_2,q_2)}, \quad 0 \le p_i \le q_i \le 1, \quad 0 \le w \le 1.$$

It is parametrized by (p_1,q_1,p_2,q_2,w).

The point of the folding is exactness of the coupling cost. If π is a symmetric coupling of μ with itself (which suffices: replacing π by $(\pi+\pi^\top)/2$ preserves both marginals and every symmetric cost), folding $(p,r) \mapsto$

$(\min(p,r),\max(p,r))$ yields a law ν on Δ whose induced marginal is μ and whose $s^\star$-cost is unchanged. Hence

$$\mathrm{C}(\mu) = \min_{\substack{\nu\in\mathcal{P}(\Delta)\\ \mu_\nu=\mu}} \int c\,d\nu, \qquad c(p,q) = h\big(s^\star(p,q)\big),$$

and the joint optimization over laws and couplings collapses to

$$\inf_{\substack{\mu\in\mathcal{P}([0,1])\\ \mathrm{mean}(\mu)\le t}} \mathrm{F}(\mu) = \inf_{\substack{\nu\in\mathcal{P}(\Delta)\\ \mathrm{mean}(\mu_\nu)\le t}} \Phi_\alpha(\nu),$$

where Φ_α is the functional stated explicitly in Theorem 3.2.

Theorem 3.2 (Theorem B$'''$; `thmB3_proof.py`)**.** *Let Δ and the map $\nu \mapsto \mu_\nu$ be as in Definition 3.1. For every $\alpha \in [0,1]$ and $t \in [0,1]$, define on the weak-$*$ compact convex set $\mathcal{P}(\Delta)$*

$$\Phi_\alpha(\nu) = (1-\alpha)\mathrm{Q}(\mu_\nu) + \alpha \int_\Delta h(s^\star(p,q))\,d\nu(p,q) - \mathrm{L}(\mu_\nu).$$

The minimum of Φ_α over $\{\nu \in \mathcal{P}(\Delta) : \mathrm{mean}(\mu_\nu) \le t\}$ is attained by a measure supported on at most two points of Δ. Its induced marginal is therefore supported on at most four points of $[0,1]$.

The proof uses the CITED Riesz, Banach–Alaoglu, Bauer, and moment-set theorems identified below. The complete argument carries their hypotheses explicitly. The executable `thmB3_proof.py` also runs sampled NUMERICAL controls, none of which is a proof step.

Proof. Step 1: compactness. Δ is a closed subset of $[0,1]^2$, hence a compact metric space. By the Riesz representation theorem (van Neerven [15], Thm. 4.2), regular Borel measures on Δ are identified with $C(\Delta)^*$, and by Banach–Alaoglu (van Neerven [15], Thm. 4.50) the closed unit ball of $C(\Delta)^*$ is weak-$*$ compact. The set $\mathcal{P}(\Delta)$ is weak-$*$ closed there (it is the intersection of closed conditions $\langle 1,\nu\rangle = 1$ and $\langle f,\nu\rangle \ge 0$ for all $f \ge 0$), so $\mathcal{P}(\Delta)$ is weak-$*$ compact. (Equivalently, Prokhorov: every family of laws on compact Δ is tight.)

Step 2: continuity. The functions

$$\begin{aligned} g(p,q) &= \frac{2-p-q}{2},\\ \ell(p,q) &= \frac{h(p)+h(q)}{2},\\ s^\star(p,q) &= \min\big(\max(\tfrac{1}{2},p,q),\min(p+q,1)\big). \end{aligned}$$

and $c = h \circ s^\star$ are continuous on Δ ($r \ln r \to 0$ gives continuity of h at the endpoints; min and max preserve continuity). The kernel W of (5) is continuous on $[0,1]^2$ and $W(0,\cdot) = W(1,\cdot) = 0$ by direct substitution, so T and W are likewise continuous. The marginal map M is continuous and linear. If $\mu_i \to \mu$ weak-$*$, then $\mu_i \otimes \mu_i \to \mu \otimes \mu$ weak-$*$: product functions $a(x)b(y)$ separate points and form a unital subalgebra, hence are uniformly dense in $C([0,1]^2)$ (Stone–Weierstrass; van Neerven [15], Thm. 2.5), and uniform approximation extends convergence to the continuous kernel $W(1-x, 1-y)$. Thus $\nu \mapsto \mathrm{E}(\mu_\nu)$ is weak-$*$ continuous, and so is Φ_α by (10).

Step 3: global attainment. The feasible set $H_t = \{\nu \in \mathcal{P}(\Delta) : \mathrm{mean}(\mu_\nu) \le t\} = \{\nu \in \mathcal{P}(\Delta) : \mathrm{B}(\mu_\nu) \ge 1 - t\}$ is the preimage of a closed ray under a continuous map, hence weak-$*$ compact, and contains $\delta_{(0,0)}$. A global minimizer $\nu^\star$ exists; put $\beta^\star = \mathrm{B}(\mu_{\nu^\star})$.

Step 4: the slice. The slice $K_\beta = \{\nu \in \mathcal{P}(\Delta) : \mathrm{B}(\mu_\nu) = \beta^\star\}$ is nonempty, convex, weak-$*$ compact, and contained in H_t, so

$$\min_{K_\beta} \Phi_\alpha = \Phi_\alpha(\nu^\star) = \min_{H_t} \Phi_\alpha.$$

On K_β the marginal mean, hence B, is fixed. By (7), $\mathrm{Q}(\mu_\nu) = 2\beta^\star \mathrm{L}(\mu_\nu) - \mathrm{E}(\mu_\nu)$, and

$$\Phi_\alpha(\nu) = \big[2(1-\alpha)\beta^\star - 1\big]\mathrm{L}(\mu_\nu) + \alpha \int_\Delta c \, d\nu - (1-\alpha)\mathrm{E}(\mu_\nu).$$

The first two terms are weak-$*$ continuous linear functionals. For $0 \le \theta \le 1$ and $\nu_0, \nu_1 \in K_\beta$, positive semidefiniteness of W gives

$$\mathrm{E}\big(\theta\mu_{\nu_0} + (1-\theta)\mu_{\nu_1}\big) \le \theta\, \mathrm{E}(\mu_{\nu_0}) + (1-\theta)\, \mathrm{E}(\mu_{\nu_1}),$$

so $-(1-\alpha)\mathrm{E}$ is concave (here the restriction $\alpha \le 1$ is used once). Hence Φ_α is continuous and concave on the compact convex K_β.

Step 5: Bauer's minimum principle. By the Bauer maximum principle (Bauer [2]; the formulation used here is Bru–de Siqueira Pedra [3], Lemma 3.3, with the reference to Bauer's original paper checked against the publisher record), a continuous convex function on a compact convex set attains its maximum at an extreme point; applied to $-\Phi_\alpha$ on K_β, there is an extreme point $\nu^{**} \in K_\beta$ with $\Phi_\alpha(\nu^{**}) = \min_{K_\beta} \Phi_\alpha$.

Step 6: extreme points of a one-moment slice. $K_\beta = \{\nu : \int g \, d\nu = \beta^\star\}$ is a moment set cut from $\mathcal{P}(\Delta)$ by *one* non-constant moment condition (plus the mass normalization, already part of $\mathcal{P}(\Delta)$). By Winkler's theorem on extreme points of moment sets [17], as reproduced in Pinelis' Theorem 12

[13], every extreme point of such a slice is supported on at most $1+1=2$ points. For completeness we give the short direct proof in this case. Suppose an extreme $\nu \in K_\beta$ has three distinct support points, and choose pairwise disjoint Borel neighborhoods A_1, A_2, A_3 of them with positive ν-mass. The three vectors

$$v_i = \Big(\nu(A_i),\ \int_{A_i} g\, d\nu\Big) \in \mathbb{R}^2$$

are linearly dependent: let $(a_1, a_2, a_3) \neq 0$ with $\sum_i a_i v_i = 0$, and put $\eta = \sum_i a_i\, \nu|_{A_i}$. Then η is a nonzero signed measure with total mass 0 and g-moment 0, so for small $\varepsilon > 0$, the two measures $\nu + \varepsilon\eta$ and $\nu - \varepsilon\eta$ (with densities $1 \pm \varepsilon a_i$ on A_i) are distinct nonnegative members of K_β whose midpoint is ν, contradicting extremality. Hence $|\operatorname{supp}(\nu^{**})| \leq 2$.

Step 7. ν^{**} has $\mathrm{B}(\mu_{\nu^{**}}) = \beta^\star$, so $\operatorname{mean}(\mu_{\nu^{**}}) = 1 - \beta^\star \leq t$, and by Step 4 it is a global minimizer. Its marginal is supported on at most four points, and ν^{**} is a two-orbit (or one-orbit) law in the five-parameter family of Definition 3.1. □

Remark 3.3 (Why two orbits, and the controls in `thmB3_proof.py`). The decisive move is freezing B (equivalently the mean) up front: by Theorem A′ that is the *only* place the mean enters Q, so the bilinear term becomes linear and the entire functional concave, leaving two moment conditions (mass and B) and hence two orbits. The executable additionally runs three explicitly labeled NUMERICAL controls: fixed-B chord tests (worst observed $\Phi(\mathrm{mid}) - \mathrm{chord} = +2.9 \times 10^{-9}$ over 3000 random segments), a merged-atom continuity check of M and E, and a control that *removes* the fixed-B premise and detects non-concavity (-0.124 observed). The binding obstruction (Section 4) uses exactly two orbits, so the theorem is tight in orbit count.

Corollary 3.4 (Five-parameter certification target)**.** *Let* $\alpha_0 = 0.0356069$ *and* $t_{\mathrm{cert}} = 0.3820660112501052$ *be the exact rationals in Theorem 1.1. Then* (1) *holds for every* $\mu \in \mathcal{P}([0,1])$ *with* $\operatorname{mean}(\mu) \leq t_{\mathrm{cert}}$ *if and only if*

$$\Phi_{\alpha_0}(\nu) \geq 0$$

for every

$$\nu = w\delta_{(p_1,q_1)} + (1-w)\delta_{(p_2,q_2)}, \qquad 0 \leq p_i \leq q_i \leq 1, \quad 0 \leq w \leq 1,$$

satisfying $\operatorname{mean}(\mu_\nu) \leq t_{\mathrm{cert}}$.

4 The margin lemma and the diagonal inequality

The functional Φ_α vanishes identically on the "sink" laws $\{(1-u)\delta_0 + u\delta_1\}$; on the way there both L and F collapse, so a raw sign test cannot terminate. The margin lemma removes this degeneracy by dividing by L.

For $x \in [0,1]$ define the diagonal density

$$\mathrm{rh}(x) = \frac{W(1-x, 1-x)}{\ln 2} = 2(1-x)h(x) - h\big((1-x)^2\big), \tag{13}$$

$$\rho(x) = \frac{\mathrm{rh}(x)}{h(x)} \qquad (0 < x < 1). \tag{14}$$

The positive-semidefinite series in Theorem 2.4 gives a Hilbert-space feature map $\mathbf{v} : [0,1] \to \ell^2$ satisfying

$$\langle \mathbf{v}(p), \mathbf{v}(q) \rangle = \frac{W(1-p, 1-q)}{\ln 2},$$
$$\|\mathbf{v}(p)\|^2 = \mathrm{rh}(p),$$
$$\mathrm{E}(\mu) = \left\| \int \mathbf{v}\, d\mu \right\|^2.$$

Lemma 4.1 (Margin lemma; `margin_lemma.py`). *Let $\alpha \in [0,1]$ and let $\mu \in \mathcal{P}([0,1])$ satisfy $\mathrm{L}(\mu) = \int h\, d\mu > 0$. With $\mathrm{C}(\mu)$ defined by the self-coupling minimum in Theorem 1.1, put*

$$\sigma(\mu) = \int \sqrt{\mathrm{rh}(p)}\, d\mu(p), \qquad \Lambda(\mu) = 2(1-\alpha)\mathrm{B}(\mu) - 1 + \alpha \frac{\mathrm{C}(\mu)}{\mathrm{L}(\mu)} - (1-\alpha)\frac{\sigma(\mu)^2}{\mathrm{L}(\mu)}.$$

Then, for $\mathrm{F}(\mu) = (1-\alpha)\mathrm{Q}(\mu) + \alpha\mathrm{C}(\mu) - \mathrm{L}(\mu)$,

$$\mathrm{F}(\mu) \geq \mathrm{L}(\mu)\Lambda(\mu). \tag{15}$$

Equality holds whenever $|\operatorname{supp}(\mu) \setminus \{0,1\}| \leq 1$.

Proof. By the triangle inequality for Bochner integrals,

$$\mathrm{E}(\mu) = \left\| \int \mathbf{v}\, d\mu \right\|^2 \leq \left(\int \|\mathbf{v}\|\, d\mu \right)^2 = \sigma(\mu)^2.$$

Substitution in (10) gives (15). If the support has at most one non-sink point, all nonzero feature vectors are collinear, so equality holds. □

Two properties make (15) usable.

1. *Bounded, never singular.* The global Arb cover in `cert2._rho_global` PROVED the cap used by the certificate. Its approximate maximizer $p^\star \approx 0.1165$ and value $\rho_{\max} \approx 0.3049467$ are NUMERICAL summaries. Since $\mathrm{rh}(x) \le \rho_{\max} h(x)$,

   $$\sigma(\mu)^2 \le \rho_{\max} \mathrm{L}(\mu),$$

   so Λ is bounded and σ^2/L cannot diverge. Along a sequence of fixed mean t collapsing to a sink, σ^2 is quadratic in the non-sink mass while L is linear, and hence

   $$\Lambda \longrightarrow 2(1-\alpha)(1-t) - 1 = \kappa_t \approx 0.19186255 > 0$$

   at the certified parameters.

2. *Tight at the obstruction.* With the exact $a^\star, b^\star$ defined above, the law $\mu^\star = a^\star \delta_1 + (1-a^\star)\delta_{b^\star}$ has exactly one non-sink support point, so

   $$\Lambda(\mu^\star) = \frac{\mathrm{F}(\mu^\star)}{\mathrm{L}(\mu^\star)}$$

   exactly. The decimal summaries $a^\star \approx 0.0788772927059232$ and $b^\star \approx 0.3294547385030370$, the 10^{-12} evaluation, and the Jensen-variant value -1.94×10^{-2} are NUMERICAL diagnostics recorded in `margin_lemma.py`; none is a proof premise.

The margin lemma is the reason the certificate can cover the near-total-sink slab $p_1, q_1 \to 0$, $q_2 \to 1$, $w \to 1$ that defeated earlier campaigns (§5.5): there L collapses while Λ stays positive.

4.1 A linear diagonal bound

The certificate's ratio rule needs a certified upper bound on ρ that is *linear and piecewise dyadic-friendly*: over any box, the maximum of $2\min(x, 1-x)$ is attained at a corner, so $\sqrt{\mathrm{rh}} \le \sqrt{2\min(x, 1-x)}$ gives an aggregate bound with no interval blow-up.

Theorem 4.2 (rh inequality; `lemma_rh_proof.py`, `lemma_rh.py`)**.** *Let $h(u) = -u\log_2 u - (1-u)\log_2(1-u)$ on $[0,1]$, with $h(0) = h(1) = 0$, and define*

$$\mathrm{rh}(x) = 2(1-x)h(x) - h\big((1-x)^2\big).$$

For every $x \in [0,1]$,

$$\mathrm{rh}(x) \le 2\min(x, 1-x). \tag{16}$$

Moreover $\mathrm{rh}(x)/x \to 2$ as $x \downarrow 0$.

Proof. For $x \in [\frac{1}{2}, 1]$, $\mathrm{rh}(x) = 2(1-x)h(x) - h((1-x)^2) \le 2(1-x)$, as $h \le 1$ and $h \ge 0$. This is the trivial side.

For $x \in [0, \frac{1}{2}]$ the relation $\mathrm{rh} \le 2x$ is $f(x) := h\big((1-x)^2\big) - 2(1-x)\,h(x) + 2x \ge 0$. The machine-checked proof (`lemma_rh_proof.py`) is built on the SymPy-exact identity

$$\ln 2 \cdot f(x) = x^2 \ln \frac{2}{x} - x(2-x)\,\ln\Big(1 - \frac{x}{2}\Big), \qquad 0 < x \le \tfrac{1}{2}, \tag{17}$$

with $f(0) = 0$ exactly. For $0 < x \le \frac{1}{2}$ both summands on the right are strictly positive ($\ln(2/x) \ge \ln 4 > 0$; $\ln(1-x/2) < 0$ negated), so $f(x) > 0$ on $(0, \frac{1}{2}]$: positivity is immediate, with no cancellation. The near-zero structure is quantified by the explicit remainder $\ln 2\, f(x) = x^2[-\ln x + \ln 2 + 1] + R(x)$ with $-x^3/2 \le R(x) \le -(15/62)\,x^3$, giving $\ln 2\, f(x) \ge (\ln 32 + \frac{31}{32})\,x^2 \approx 4.434\,x^2$ on $(0, \frac{1}{16}]$; in particular $f(x) = x^2 \log_2(1/x) + O(x^2) = o(x)$, which is the tightness claim. On $[\frac{1}{16}, \frac{1}{2}]$ an adaptive Arb interval cover certifies $f > 0$ cell by cell; the machine proof executes a 44-subinterval cover with minimum certified lower bound 2.4×10^{-3}, after an exact recursive split of the interval. The PROVED gate checks the SymPy identity, series through order x^6, leading behavior, algebraic factorizations, and remainder bounds; the adaptive Arb cover checks exact contiguity and positive lower bounds. The executable prints `PROOF COMPLETE: machine-checked` only after both gates pass. □

A companion monotonicity fact supplies the endpoint-safe caps used by `cert2.rho_upper`:

$$\frac{d}{dz}\frac{z}{h(z)} = \frac{h(z) - z\,h'(z)}{h(z)^2} = \frac{-\log_2(1-z)}{h(z)^2} > 0 \qquad (0 < z < 1), \tag{18}$$

so $z/h(z)$ increases and by symmetry $h(z) = h(1-z)$ the ratio $(1-z)/h(z)$ decreases. These signs are proved symbolically by `cert2._RHO_NUMERATOR_IDENTITY` and asserted at import by `prove_rho_endpoint_monotonicity`.

5 The certificate

5.1 The reduced problem and exact arithmetic

By Corollary 3.4 it remains to certify, for $\alpha = 0.0356069$ and $t_{\mathrm{cert}} = 0.3820660112501052$, that $\Phi_\alpha(\nu) \ge 0$ for every two-orbit $\nu = w\,\delta_{(p_1,q_1)} +$

$(1-w)\,\delta_{(p_2,q_2)}$ with $\mathrm{mean}(\mu_\nu) \le t_{\mathrm{cert}}$. All universal inequalities are statements in *Arb ball arithmetic* (`flint/Arb`, Johansson [8]): enclosures of h, $s^\star$ (monotone corners), rh, and $\sqrt{\mathrm{rh}}$; sound min/max extensions; exact shortest-decimal parsing of float endpoints into Arb; certified covers. Every rule is an independently sound lower bound or infeasibility proof: *skipping a rule can only send a box to the split branch, never clear it.* Numerically derived quantities that steer the search are labeled NUMERICAL and are never used as proof steps.

The exact orbit swap $(p_1, q_1, p_2, q_2, w) \mapsto (p_2, q_2, p_1, q_1, 1-w)$ preserves $\mathrm{mean}, \mathrm{L}, \mathrm{C}, \mathrm{E}, \Phi_\alpha$ term by term (asserted exactly and checked on rational endpoints by `verify_orbit_swap`), so the root $[0,1]^4 \times [\frac{1}{2}, 1]$ covers the full cube, and it is split into the eight exact dyadic w-slices $[\frac{1}{2}, \frac{9}{16}], [\frac{9}{16}, \frac{5}{8}], \ldots, [\frac{15}{16}, 1]$ (`root_w_lo = 1/2`, `root_w_hi = 1` in the manifest). Campaign parameters (v2 manifests): $\alpha = 0.0356069$, $t = 0.3820660112501052$, `level_offset` $= 10^{-4}$, min box width 10^{-3}, collar floor 1.25×10^{-4}, face floor 10^{-3}, face box budget 10^5, box budget 2×10^9, 24 h per slice (72 h in the relaunched campaign I), 160-bit native covers with an 80-bit working precision, `center_gate` $= -0.02$, `centered5_max_width` $= 1/32$, `q2pin` $= 15/16$, `collar_edge` $= 0.01$.

λ-shifts. For $\lambda \ge 0$, the shifted functional $\Phi_{\alpha,\lambda} = \Phi_\alpha + \lambda(\mathrm{mean} - t)$ satisfies $\Phi_{\alpha,\lambda} \le \Phi_\alpha$ on feasible points ($\mathrm{mean} \le t$), so any certified lower bound on the shifted functional over a box is a valid lower bound on Φ_α there. The certified family is

$$\lambda_{\mathrm{seed}} = 1.3682158100251758,$$
$$\mathcal{L} = \{\, 0,\ 1,\ 1.626,\ 2.2,\ 0.98\lambda_{\mathrm{seed}},\ \lambda_{\mathrm{seed}},\ 1.02\lambda_{\mathrm{seed}} \,\}.$$

The displayed decimal is an exact rational seed in the manifest, chosen from a NUMERICAL face-KKT solve (reported residual below 10^{-35}). Correctness does not depend on the seed satisfying any KKT equation.

5.2 The rule chain

`cert3.certify` runs the following cheap-first cascade on a stack of boxes; each rule writes one trace byte (codes in §5.3).

0 *Mean contractor* (`diag_exhaust.mean_contract`); with exact rationals $l_i = (p_i^{\mathrm{lo}} + q_i^{\mathrm{lo}})/2$ from shortest-decimal endpoints, every point of the box satisfies $\mathrm{mean} \ge A(w) := l_2 + w(l_1 - l_2)$; solve $A(w) \le t$ exactly in `Fraction` arithmetic to contract w or prove the box infeasible (trace code 0).

1,2 *Corner bound* (`diag_exhaust.phi_corner`). With $m_\star = \min(\mathrm{mean}_{\mathrm{hi}}, t)$ and $\kappa_\star = 2(1-\alpha)(1-m_\star) - 1$ required > 0, then $\Phi \geq \kappa_\star \, \mathrm{L}_{\mathrm{lo}} + \alpha \, \mathrm{C}_{\mathrm{lo}} - (1-\alpha)\, \sigma^2_{\mathrm{hi}}$ over the box (code 1 = infeasible, code 2 = corner-clear).

3 *Ratio rule* (`cert2.ratio_rule`). From the margin lemma, with a *box-specific* w-window $W = \mathtt{hull}(w_{\mathrm{lo}}, w_{\mathrm{hi}})$: $\sigma^2 \leq \mathrm{L} \int \rho \, d\nu \leq \mathrm{L}\, \rho_{\mathrm{hi}}$, so

$$\Phi \geq \mathrm{L}\big[\kappa_t - (1-\alpha)\rho_{\mathrm{hi}}\big],$$
$$\kappa_t = 2(1-\alpha)(1-t) - 1 \approx 0.191862550011752.$$

with ρ_{hi} the endpoint maximum over W of the convex combination of the two orbits' certified caps (the combination is linear in w; an unsound slope-form co-variation was identified and replaced by the endpoint max). This rule is *its own cheapest applicability test* (35.6 μs vs. 27.5 μs for the corner bound): a hand-tuned geometric pre-filter ("every atom ≤ 0.05 or ≥ 0.42") delegating applicability was removed because it fell silent exactly on the near-total-sink slab that matters; see §5.5.

4--7 *Centered (KKT-shifted) rules.* The primary rule is `centered5_best`, with `centered5_mixed`, `centered5_mixed_swap`, and `centered_w_best` as block-mixed, orbit-swap, and weight-only fallbacks. They are gated to widths $\leq 1/32$ and require a corner lower bound at least -0.02. Each proves the lower bound $\Phi_\lambda \geq \Phi_\lambda(\mathrm{mid}) - \sum_i \sup_{\mathrm{box}} |\partial_i \Phi_\lambda| \cdot w_i$ built from certified interval gradients over the λ-family, falling back to block-mixed forms when the full gradient is unusable at a sink-touching vestigial orbit, and to the exact orbit-swapped form or the derivative-free weight-only form on collar-touching boxes.

8 *q_2-pin face rule.* When $q_2^{\mathrm{lo}} \geq \mathtt{q2pin} = 15/16$, $s^\star(p_2, q_2) = q_2$ is pinned and $\partial \Phi_\lambda / \partial q_2 \leq 0$ on the strip, so the q_2-face carries the minimum; a four-dimensional face branch-and-bound (`face_bb`) certifies that face with the pinned λ-subfamily.

9 *Residual.* All five coordinates at their floors (width $\leq 10^{-3}$, or $\leq 1.25 \times 10^{-4}$ for collar-touching boxes): an uncleared box, recorded with its center and full widths. A COMPLETE slice has zero residuals.

16+j *Split of coordinate j.* The splitter chooses the coordinate of largest *influence-weighted* width (width $\times$ influence, where orbit 1's atoms carry at most w^{hi} and orbit 2's at most $1 - w^{\mathrm{lo}}$; `split_by_influence`, with `split_by_width` kept beside it behind the `SPLIT_CHOICE` hook).

Any split is sound (children cover the parent exactly); the choice affects only cost.

A box is *terminal* when a rule proves infeasibility, certifies $\Phi \geq 0$, or cannot be split. The verdict is `COMPLETE` iff the stack is empty, residuals are zero, and the box and time budgets were not consumed.

5.3 Proof traces and independent replay

Every processed raw DFS node writes exactly one byte (`cert3-trace-v1`): codes 0–9 name the terminal rule, code $16 + j$ the split coordinate. The trace size therefore *equals* the processed tally by construction.

`cert3_replay.replay_trace` reconstructs every box from the exact dyadic root by re-applying the mean contractor and the recorded splits (strict, representable midpoints required), re-proves *only the claimed rule* per terminal in Arb interval arithmetic (splits are intrinsically sound), rejects any residual event, unknown code, trailing event, or non-empty final stack, and compares all trace-derived tallies against the committed record. Worker fidelity is enforced by import order: same modules, 160-bit covers, then 80-bit working precision; the λ-family is recomputed and checked against the manifest. A replay failure is a rejection problem.

- *Adversarial suite*: all eight attack vectors are verified rejections — flipped rule byte; injected residual; clear-replaced-by-split; truncation; extension; unknown code; false infeasibility claim; wrong expected tallies — each with the correct diagnostic. An INCOMPLETE slice trace replays mathematically and is rejected precisely for its pending stack.
- *Determinism*: campaigns E and F ran slice 0 five hours apart on different snapshots; both processed exactly 16,231,621 boxes and their trace files are byte-identical (SHA-256 `9eed0cc1382f5485...`). The proof object is reproducible, not merely re-derivable.

5.4 The immutable campaign protocol (v2)

`cert3_par.py create` writes a fresh UUID-named campaign directory. The seven executable sources are `cert3.py`, `cert2.py`, `cert3_par.py`, `diag_exhaust.py`, `bound_kkt.py`, `arbcore.py`, and `entropy.py`. The eight review sources are `bridge_uc.py`, `cert3_collect.py`, `cert3_replay.py`, `decomposition.py`, `lemma_rh_proof.py`, `margin_lemma.py`, `reduction.`

py, and `thmB3_proof.py`. The protocol copies exactly those 15 files into `snapshot/`, hashes each with SHA-256, and atomically writes `launch.json` with exact dyadic roots, canonical lowercase-hex UUID4 run identifiers, and exact parameters. Workers run from the snapshot under `python -B`; the exact listing prevents imports from preferring stray bytecode or extension modules.

Commit is atomic and append-only. The worker writes its trace to a run-unique hidden temp file with flush+fsync, records its SHA-256 and size (size must equal processed), and the supervisor re-hashes the temp before installing it next to the result via Darwin `renamex_np` with `RENAME_EXCL`: nothing is ever deleted or overwritten, and two writers cannot collide.

`cert3_collect.py` accepts a campaign only by *re-proving it.* Its hard-coded `EXPECTED_EXECUTABLE_FILES` (7) and `EXPECTED_REVIEW_FILES` (8) override the manifest's inventory; the snapshot is re-listed, re-hashed, and must be pristine; roots must be the exact dyadic partition, run ids canonical UUID4, exit code 0, verdicts COMPLETE, positive work, and `stack` = `residual` = `budget_boxes` = `budget_time` = 0; trace digests and sizes consistent; then it re-imports the frozen modules from the snapshot (import location asserted) and *replays all eight traces* before printing `COMPOSITE CERTIFICATE`. Hand-authored complete JSON is insufficient by construction.

5.5 The campaign record: E, F, G, H, I, and J status

The certified rational decimal $t_{\mathrm{cert}} = 0.3820660112501052 \geq \psi + 10^{-4}$ is itself proven exactly: for $r \in [0, \frac{3}{2}]$, $r \geq \psi \iff r^2 - 3r + 1 \leq 0$, checked in rationals (`cert3_par.target_relation_holds`).

How the residual barrier was found and removed (E,F,G). Campaign E's slice 6–7 incompleteness was purely wall-clock (12 and 18 pending stack boxes at the 8 h wall, zero residuals). Campaign F's slice 7 exposed a genuinely mathematical-looking wall: 1,447,470 residuals in 51.4 M boxes, all `sink`, pinned to the razor-thin near-total-sink slab $p_1, q_1 \in [0, 10^{-4}]$, $q_2 \in [0.9844, 1]$, $w \in [0.9920, 1]$, $p_2 \in [0.4063, 0.4200]$ — the degenerate family the margin lemma was built for; the ratio rule cleared 400/400 sampled leaves but was *never called* there, as a hand-tuned filter demanded every atom ≤ 0.05 or ≥ 0.42 and p_2 missed 0.42 by 1.4×10^{-2}. Removing the filter is sound (a skipped bound only forces a split), the rule is its own cheapest test, and the extracted slab certifies COMPLETE in 36,511 boxes. Campaign G ran without it: trees *shrunk* (15.2–29.9 M vs. 16.2–33.6 M) while

slice	E proc.	F proc.	G proc.	G ratio	H proc.	I proc.
0	16,231,621	16,231,621	15,528,149	192,277	21,135,611	21,135,611‡
1	16,320,717	16,320,717	15,226,435	228,386	20,827,065	20,827,065‡
2	17,058,935	17,058,935	15,381,807	295,807	23,486,265	23,486,265‡
3	19,213,165	19,213,165	17,831,917	281,389	34,186,855	34,186,855‡
4	19,341,141	19,341,141	19,030,155	82,944	54,161,482†	71,405,987*
5	22,972,035	22,972,035	22,640,283	78,913	59,867,104†	117,709,435*
6	21,467,895	33,614,993	29,889,721	271,760	74,343,239†	121,456,775*
7	26,095,385	51,360,695	52,086,331	1,193,438	78,257,861	78,257,861‡

Table 1: Processed boxes and (for G) ratio clears per w-slice in campaigns E, F, G, H, and I; residuals are 0 except F slice 7 (1,447,470 sink residuals), F slices 6–7 and G slice 7 stopped by the wall (stacks 12/18, 24), G slice 7 at 52,086,331 boxes with residual 0. E: 2026-08-15, 8 h/slice, slices 0–5 COMPLETE and replayed. F: 2026-08-16 05:38, 12 h/slice, slices 0–6 COMPLETE and replayed. G: 2026-08-16 17:56, 24 h/slice, slices 0–6 COMPLETE and replayed, all rows summing to the committed 187,614,798 processed nodes. H (final): 2026-08-17 21:20, 24 h/slice, slices 0–3 and 7 COMPLETE and replayed, rows summing to the committed 366,265,482 processed boxes with residual 0 throughout; † marks the INCOMPLETE slices 4–6, stopped by the 24 h wall with stacks 25/27/30 and residual 0 (time-limited only). I (relaunch): all eight slice records are committed, rows summing to 488,465,854 boxes. ‡ marks "trace SHA-256 identical to campaign H" (slices 0–3, 7); $\star$ marks "I total exceeds H wall" (slices 4–6). The frozen collector replayed all eight traces in its own gate and printed `COMPOSITE CERTIFICATE` on 2026-08-21. On every replayed slice the binary-tree invariant terminals = splits + 1 holds exactly.

ratio clears rose (3.7× on slice 0); slice 7 passed F's residual onset at box 26.1 M at residual 0 and reached 52,086,331 boxes before the 24 h wall.

Campaign H's only change is the splitter. The NUMERICAL cost probe classified slice-7 splits as distributed (48.5 % `other`, 35.1 % `mean-boundary`, 16.4 % `sink`), so the remaining cost appeared computational rather than a new inequality; `split_by_influence` weights a coordinate's width by its orbit's weight cap (on slice 7, orbit 2's coordinates are discounted by up to 16×). The measured changes — the former residual slab 36,511 → 131 boxes, the tight-obstruction root 231 → 109 boxes, and 2.8× throughput on $w \in [0.99, 1]$ — are all NUMERICAL. Campaign H's mathematical acceptance is by replay alone.

Campaign H outcome; campaign I. H finished on 2026-08-18: slices 0–3 and 7 COMPLETE — 21,135,611, 20,827,065, 23,486,265, 34,186,855 and 78,257,861 boxes, stacks and budget flags all zero — and slices 4–6 INCOMPLETE at the wall only, at 54,161,482, 59,867,104 and 74,343,239 processed boxes with stacks 25/27/30. Slice 7, which had defeated E (budget), F (residuals) and G (budget), finished with 7 h of headroom; its 78,257,861-event trace replayed independently in 2 h 11 m (PASS), as did the four smaller echoes, every tally matching the committed records and terminals = splits + 1 exact. Across all 366,265,482 boxes the residual count never left zero: the three remaining slices wanted hours, not mathematics. Campaign I (2026-08-18 21:26 UTC, workers `certI0--7`) committed all eight slices under the same code hash and exact target with a 72 h budget per slice. Its totals are 21,135,611, 20,827,065, 23,486,265, 34,186,855, 71,405,987, 117,709,435, 121,456,775 and 78,257,861, for 488,465,854 boxes overall. The I traces on slices 0–3 and 7 are byte-identical to H by SHA-256; slices 4–6 have totals exceeding H's wall totals. The frozen collector replayed all eight traces and printed `COMPOSITE CERTIFICATE` on 2026-08-21: campaign I certifies the rational target $t_{\mathrm{cert}} = 0.3820660112501052 \geq \psi + 10^{-4}$.

Second campaign (closed; no certificate). Campaign J, directory `cert3_20260820T181524Z_08ad738a1cf24a08bfc4189e65f9ef44_57a8908ba7b8` with code hash `57a8908ba7b8`, ran with a 72-hour budget per slice at the rational target $t_J = 0.3821660112501052 \geq \psi + 2 \cdot 10^{-4}$. Six slices committed `COMPLETE` with residual and stack zero: slices 0–4 and 7 processed 21,891,091, 21,637,061, 24,563,611, 37,877,105, 85,793,971, and 93,281,643 boxes. Slices 5 and 6 reached 142,886,584 and 156,713,996 boxes with residual zero, but hit the 72-hour wall with stack and `budget_time` tallies 31 and 29. The campaign processed 584,645,062 boxes with no residual leaf. Its frozen collector exited 1 at admission, without replay, and printed `NO COMPOSITE CERTIFICATE` with exactly six problems: non-COMPLETE verdict, nonzero stack, and nonzero time budget for each of those two slices. Thus campaign J provides deterministic cost evidence and a negative-control collector run, but no certification claim at t_J.

6 A proof of Liu's positive-semidefiniteness hypothesis

Liu's conditionally iid argument [10, Theorem 13 and §V] has two separate conditional inputs. Its Hypothesis 1, in §V-A, is a positive-semidefiniteness assertion; Hypothesis 2, in §V-B, asserts the structure of a nine-parameter global optimizer. Assuming both, Liu obtains the union-closed constant

$$0.382709087918741 > c^\star = 0.3823455333667027,$$

strictly beyond the ceiling of the two-strategy method used by our certificates. The result below proves Hypothesis 1. Hypothesis 2 is not addressed and remains open. Liu's numerical check of Hypothesis 1 used a float64 eigenvalue computation on a 2499-point grid and reported a largest eigenvalue 1.6311×10^{-14}, below the roundoff scale of a dense computation of that size.

For this section put

$$a_{\mathrm{L}}(s) = s(1-s), \qquad z(s,t) = (1-s)(1-t) + a_{\mathrm{L}}(s)a_{\mathrm{L}}(t).$$

Liu's hypothesis says that

$$\iint h(z(s,t))\, d\mu(s)\, d\mu(t) \le 0 \tag{19}$$

for every finite signed measure μ annihilating

$$1, \qquad s, \qquad a_{\mathrm{L}}(s) = s(1-s).$$

These three functions span all polynomials of degree at most two. Thus the hypothesis has codimension three, as does Liu's projector $[1, i/n, a_{\mathrm{L}}(i/n)]$; the positive-semidefiniteness terminology refers to the negative of the entropy kernel on this subspace.

6.1 Exact reduction to an unprojected kernel

Set

$$u = 1-s, \qquad v = 1-t, \qquad A = uv, \qquad B = st,$$

so that $a_{\mathrm{L}}(s)a_{\mathrm{L}}(t) = AB$ and $z = A(1+B)$. With

$$G(x) = x + (1-x)\ln(1-x) = \sum_{k=2}^{\infty} \frac{x^k}{k(k-1)}$$

on $[0,1]$, the entropy has the exact expansion

$$\begin{aligned} \ln 2\, h(z) &= -z\ln u - z\ln v - z\ln(1+B) + z - G(z) \\ &= -z\ln u - z\ln v - z\ln(1+B) + z - \sum_{k=2}^{\infty} \frac{z^k}{k(k-1)}. \end{aligned} \tag{20}$$

Products involving an endpoint logarithm are understood by continuous extension. The three displayed terms $-z\ln u$, $-z\ln v$, and $+z$ are exactly the pieces discarded by Liu's projection. Indeed,

$$\begin{aligned} -z\ln u &= -(u\ln u)v - (us\ln u)(vt), \\ -z\ln v &= -u(v\ln v) - (us)(vt\ln v), \\ z &= uv + (us)(vt), \end{aligned}$$

and every rank-one summand has, in at least one variable, a polynomial factor of degree at most two. This is precisely why all three constraints $1, s, a_{\mathrm{L}}(s)$ are needed.

The rank-one kernel $AB = a_{\mathrm{L}}(s)a_{\mathrm{L}}(t)$ is itself annihilated. Moving that null summand from $z = A + AB$ into the remainder rewrites (20) as

$$\ln 2\, h(z) = -z\ln u - z\ln v + A + R(s,t),$$

where

$$R(s,t) = A\bigl[B - (1+B)\ln(1+B)\bigr] - \bigl[z + (1-z)\ln(1-z)\bigr]. \tag{21}$$

Consequently, for every μ satisfying Liu's three constraints,

$$\ln 2 \iint h(z(s,t))\, d\mu(s)\, d\mu(t) = \iint R(s,t)\, d\mu(s)\, d\mu(t). \tag{22}$$

Thus it suffices to prove the stronger statement that R is negative semidefinite without any projection.

Theorem 6.1 (Liu's positive-semidefiniteness hypothesis)**.** *The continuous kernel R in (21) is negative semidefinite on $[0,1]$: for every $\nu \in \mathcal{M}([0,1])$,*

$$\iint R(s,t)\, d\nu(s)\, d\nu(t) \le 0.$$

In particular, its integral operator is negative semidefinite on $L^2[0,1]$, and (19) holds.

Proof. Continue to regard $A = uv$ and $B = st$ as kernels. Extend $G(x) = x + (1-x)\ln(1-x)$ to $[-1,1]$. Since $G(-B) = -B + (1+B)\ln(1+B)$, define

$$\phi(B) = G\bigl(A(1+B)\bigr) + A\,G(-B) = -R.$$

Treating A as fixed, direct symbolic differentiation gives

$$\phi(0) = G(A), \qquad \phi'(0) = -A\ln(1-A),$$
$$\phi''(B) = \frac{A}{(1+B)\,[1-A(1+B)]}.$$

Taylor's theorem with integral remainder therefore gives the exact identity

$$\begin{aligned} -R = G(A) - AB\ln(1-A) \\ + B^2 \int_0^1 (1-r)\frac{A}{(1+rB)\,[1-A(1+rB)]}\,dr. \end{aligned} \tag{23}$$

The first two kernels in (23) are positive semidefinite by nonnegative power series and the Schur product theorem:

$$G(A) = \sum_{k=2}^{\infty} \frac{A^k}{k(k-1)}, \qquad -AB\ln(1-A) = \sum_{k=1}^{\infty} \frac{A^{k+1}B}{k}.$$

Indeed, $A^k = u^k v^k$ and $A^{k+1}B = (u^{k+1}s)(v^{k+1}t)$ are rank-one positive-semidefinite kernels.

It remains to treat the integral remainder. For $0 \le r \le 1$ write

$$D_r(s,t) = (1+rB)\,[1-A(1+rB)].$$

Let $\mathbf{m}(x) = (1, x, x^2, x^3)^\mathsf{T}$. In this polynomial basis the kernel has coefficient matrix

$$\begin{aligned} M(r) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & -r-1 & 2r & 0 \\ 0 & 2r & -r(r+2) & r^2 \\ 0 & 0 & r^2 & -r^2 \end{pmatrix}, \\ D_r(s,t) = \mathbf{m}(s)^\mathsf{T} M(r)\mathbf{m}(t). \end{aligned} \tag{24}$$

Exact coefficient extraction gives

$$\det M(r) = -2r^3,$$

and the coefficients of $\det(\lambda I - M(r))$, in descending powers of λ, are

$$1, \quad (r+1)(2r+1), \quad 4r^3+2r-1, \quad -2r(r^3-r^2+r+1), \quad -2r^3. \tag{25}$$

For $0 < r \le 1$ their signs are $(+,+,\pm,-,-)$, with a zero coefficient omitted. There is exactly one Descartes sign change. Since $M(r)$ is symmetric and its determinant is nonzero, it has inertia $(1\ \text{positive}, 3\ \text{negative})$.

For completeness, an exact LDL^{T} factorization makes the resulting Lorentz decomposition explicit. Put

$$\zeta_0(s) = -\frac{2rs^2 - (r+1)s + 1}{r+1}, \quad \zeta_1(s) = 1 + 2rs^2,$$
$$\zeta_2(s) = \frac{s^2(r+2-rs)}{r+2}, \qquad \zeta_3(s) = s^3,$$

and

$$\alpha_r(s) = \frac{\zeta_1(s)}{\sqrt{r+1}},$$
$$\mathbf{b}_r(s) = \left(\sqrt{r+1}\,\zeta_0(s),\ \sqrt{r(r+2)}\,\zeta_2(s),\ r\sqrt{\frac{2}{r+2}}\,\zeta_3(s)\right).$$

Then, also at $r = 0$ by continuity,

$$D_r(s,t) = \alpha_r(s)\alpha_r(t) - \langle \mathbf{b}_r(s), \mathbf{b}_r(t)\rangle. \tag{26}$$

The positive coordinate cannot degenerate away from the corner. On the diagonal,

$$\begin{aligned} D_r(s,s) &= (1+rs^2)\,[1-(1-s)^2(1+rs^2)] \\ &\geq s(2-2s+2s^2-s^3) > 0 \qquad (0 < s \leq 1). \end{aligned} \tag{27}$$

For the inequality, apply concavity of $q \mapsto q[1-(1-s)^2 q]$ on $1 \leq q \leq 1+s^2$ and check its two endpoints. The cubic $p(s) = 2-2s+2s^2-s^3$ has $p'(s) = -3s^2+4s-2$, whose discriminant is -8; hence p decreases from 2 to 1 on $[0,1]$.

Set $\mathbf{c}_r(s) = \mathbf{b}_r(s)/\alpha_r(s)$. Equation (26) and (27) give $\|\mathbf{c}_r(s)\| < 1$ for $s > 0$. Thus Cauchy–Schwarz gives absolute convergence of the Gram series

$$\frac{1}{D_r(s,t)} = \frac{1}{\alpha_r(s)\alpha_r(t)} \sum_{k=0}^{\infty} \left\langle \mathbf{c}_r(s)^{\otimes k}, \mathbf{c}_r(t)^{\otimes k} \right\rangle \qquad (s,t>0). \tag{28}$$

Every summand is a positive-semidefinite kernel. Multiplication by the rank-one kernel

$$AB^2 = (1-s)s^2(1-t)t^2$$

therefore shows that AB^2/D_r is positive semidefinite on $(0,1]^2$. It extends continuously by zero to the closed square: near $(0,0)$, $D_r(s,t) = s+t+O((s+t)^2)$ uniformly for $r \in [0,1]$, whereas $AB^2 = O(s^2t^2)$. Positive semidefiniteness at boundary points follows by taking limits of the finite

Gram matrices. Integration against the nonnegative weight $(1-r)\,dr$ preserves positive semidefiniteness.

The middle term of (23) also has the asserted continuous extension at the only singular corner. If $m = \max(s,t)$, then $1 - A = s + t - st \geq m$ and $B = st \leq m^2$, so

$$0 \leq AB[-\ln(1-A)] \leq m^2(-\ln m) \longrightarrow 0.$$

Hence every term in (23) extends to $[0,1]^2$ as a positive-semidefinite kernel. Their sum is $-R$, so R is negative semidefinite. The signed-measure statement follows either from the displayed Gram representations or by approximating against the continuous kernels, and (22) proves Liu's Hypothesis 1. $\square$

Scope. Theorem 6.1 settles Liu's Hypothesis 1 only. Hypothesis 2, the nine-parameter global-optimization assertion in Liu's §V-B, remains open. Consequently the value 0.382709087918741 remains a conditional union-closed constant; this paper makes no unconditional claim at that value.

Remark 6.2 (The printed constant is a rounding of the equation-defined root). Liu's defining equations §V-B (90)–(93) determine

$$c' = 0.38270908791873502993031290 21\ldots, \qquad \beta^* = 0.10005255986289310666462 83\ldots,$$

while the paper prints the 15-digit values 0.382709087918741 and 0.100052559862974. Taking the printed pair literally as exact rationals, one finds an exact-rational point with the mean constraint active and diagonal representation whose objective is enclosed in interval arithmetic strictly below 1, namely 0.9999999999999903285961768504459…, so the printed decimal overshoots the root by roughly 6×10^{-15} and the inequality *as printed* fails. This is an artifact of rounding the last digit upward, not a defect in Liu's argument: the equation-defined constant is unaffected. The practical consequence is that any certificate along this route must target a rational at or below 0.38270908791873503, and comparisons with $c^\star$ should use the equation-defined value.

Remark 6.3 (Why the hypothesis resisted, and tightness). Three natural proof routes can be ruled out. First, termwise Cauchy–Schwarz/telescoping domination cannot close: the negative double-series family is confined to the index band $i \leq k$, while every positive term is of the form us^j with $j \geq 3$, so the available negative weight at the first positive index is exactly zero. Second, a uniform spectral-gap comparison cannot close because the negative-side operator is compact on an infinite-dimensional space and has lower spectral edge zero. Third, a Schur factorization through $(1+\theta st)^{-1}$ is invalid because that kernel is not positive semidefinite. Already at $\theta = 1/2$ and points $1/4, 3/4$, its 2×2 Gram determinant is the exact negative rational $-131072/1657425$; larger NUMERICAL Gram tests had minimum eigenvalues from -1.02 at $\theta = 0.25$ to -9.47 at $\theta = 1$ for 60 and 200 nodes. The inequality is tight at the corner: the resolved NUMERICAL generalized ratio is 0.9999999906, and the factorization identifies the exact source of the lost margin as $D_r(0,0) = 0$.

Corroboration only (not part of the proof). The symbolic identities and diagnostics are implemented in `liu_kernel.py`, `liu_R_nsd.py`, `liu_tail.py`, and `liu_moment_cert.py`. On a 300-node grid, (23) reproduced $-R$ to 8.9×10^{-16}; the three pieces in its displayed order had minimum eigenvalues -1.09×10^{-14}, -1.92×10^{-15}, and -4.78×10^{-16}. The sampled kernels A/D_r had minimum eigenvalue about -7×10^{-14} for $r \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}$. Independently, exact moments followed by Arb arithmetic certified the following upper bounds for compressions of R to polynomials of degree at most D, in the orthonormal shifted-Legendre basis $\sqrt{2n+1}\,P_n(2s-1)$:

$$\begin{aligned} D = 8: &\quad \lambda_{\max} \le -1.6433552396584593498213 0 \times 10^{-10}, \\ D = 12: &\quad \lambda_{\max} \le -1.3930180656209241519160 0 \times 10^{-14}, \\ D = 16: &\quad \lambda_{\max} \le -1.3761442120749068466119 9 \times 10^{-18}. \end{aligned}$$

These are strictly negative validated enclosures on the stated finite subspaces. They corroborate, but do not replace, the continuum proof above.

A Reproducibility

A.1 Frozen artifact and collector command

The certified campaign directory is

```
cert3_20260818T212601Z_425f109c15b64a6198785c6cebbbdaab_2f
23a58ebdb8.
```

From the repository's `math/` directory, the complete replay is the following single command:

```
.venv/bin/python -B uc/campaigns/cert3_20260818T212601Z_425f109c
15b64a6198785c6cebbbdaab_2f23a58ebdb8/snapshot/cert3_collect.py
```

uc/campaigns/cert3_20260818T212601Z_425f109c15b64a6198785c6cebbb
daab_2f23a58ebdb8/launch.json

Exit code 0 and the final line `COMPOSITE CERTIFICATE` are jointly required. The accepted collector has SHA-256

95d09322a7821f41850ef1ce77345e5e93eb4fa570cc0e7f58f817716c
f681ec.

The full evidence bundle used for independent transfer is

/Users/jinleic/jinleic-workspace/data/uc-certificate-psi
-plus-1e-4-2026-08-21-v2.zip.

A.2 Pinned environment

The launch manifest pins CPython 3.14.3, flint 0.9.0, mpmath 1.3.0, numpy 2.5.2, scipy 1.18.0, and sympy 1.14.0 on macOS arm64. Workers and the collector run the frozen sources with `python -B`. The accepted code SHA-256 is `2f23a58ebdb8b14275284e2ffefca7862373f7baf5d354a06e607cff8d78ce05`. Never import the snapshot modules from another program: importing can create a bytecode cache before the module disables bytecode writes, thereby changing the immutable snapshot listing.

A.3 Committed tallies and replay time

Every campaign-I slice was committed with

$$\texttt{stack} = \texttt{residual} = \texttt{budget\_boxes} = \texttt{budget\_time} = 0.$$

The collector record is:

slice	processed boxes	replay elapsed (s)
0	21,135,611	2,436.6
1	20,827,065	2,442.1
2	23,486,265	2,721.5
3	34,186,855	4,326.3
4	71,405,987	9,746.8
5	117,709,435	25,329.8
6	121,456,775	21,622.3
7	78,257,861	9,115.2
total	488,465,854	77,740.6

Thus the practical expected serial replay wall time on the pinned host is about 77,741 seconds, or 21 hours 36 minutes. Hardware and system load can change elapsed time but cannot change any accepted tally or verdict.

A.4 Acceptance checks and residual trust base

The collector performs the protocol of §5.4: it checks its hardcoded source inventories, pristine snapshot listing and hashes, exact dyadic roots, unique UUID4 run identifiers, worker exit codes, COMPLETE verdicts, positive work, zero stack/residual/budget fields, and trace digest/size consistency. It then imports only the frozen modules and mathematically replays every claimed discharge in Arb arithmetic. A hand-authored result JSON cannot pass this gate.

The residual trust base is explicit: CPython semantics and startup behavior; the pinned flint/Arb implementation; mpmath, numpy, scipy, and sympy at the versions above; Darwin `renamex_np` with `RENAME_EXCL`; SHA-256; the hashed Python sources; and the CITED results used in the analytic reduction — Riesz, Banach–Alaoglu, and Stone–Weierstrass as presented by van Neerven [15], Bauer's principle [2, 3], the moment-set theorem [17, 13], and Cambie's single entropy-to-union-closed implication [5, Question 2 and §4].

Artifact map. `bridge_uc.py` checks the $s^\star$ identity and folding bridge; `decomposition.py` checks Theorem 2.2; `reduction.py` checks Theorem 2.4, Corollary 2.5, Corollary 2.6, and Lemma 2.7; `thmB3_proof.py` checks Theorem 3.2; `margin_lemma.py` checks Lemma 4.1; `lemma_rh_proof.py`, `cert2.py`, and `arbcore.py` check Theorem 4.2 and its caps; `cert3.py` writes the certificate and traces; `cert3_replay.py` replays them; `cert3_par.py` enforces the campaign protocol; and `cert3_collect.py` performs final acceptance. `liu_kernel.py` checks the codimension-three reduction; `liu_R_nsd.py` checks the Taylor and Lorentz identities and the Arb compressions; and `liu_tail.py` and `liu_moment_cert.py` audit the failed routes and corroborating calculations.

Acknowledgments

This note is part of the union-closed certification project. The companion proof ledger and status documents are maintained with the repository artifacts. All certified computations reported here were performed on the pinned macOS arm64 environment described in Appendix A.

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